Lab tutorial 2

Exercise 1.
A student is to answer 7 out of 10 questions in an examination.

(i)- How many choices she has?

(ii)- How many if she must answer at least 3 of the first 5 questions?

Solution 1.

(i)- This is a combination problem, thus she has

$$\binom{10}{7} = \frac{10!}{7!(10-7)!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$

choices.

(ii)- For 3 questions of the first 5, she has

$$\binom{5}{3}\binom{10-5}{7-3} = \frac{5!}{3!(5-3)!} \cdot \frac{5!}{4!(5-4)!} = 10 \cdot 5 = 50$$

choices. And for 4 questions of the first 5, she has

$$\binom{5}{4}\binom{10-5}{7-4} = 5 \cdot 10 = 50$$

choices. And for 5 questions of the first 5, she has

$$\binom{5}{5}\binom{10-5}{7-5} = 1 \cdot 10 = 10$$

choices. Finally, she has $50 + 50 + 10 = 110$.

Exercise 2.
A police department in a small city consists of 10 officers. If the department policy is to have 5 of the officers patrolling the streets, 2 of the officers working full time at the station, and 3 of the officers on reserve at the station, how many different divisions of the 10 officers into the 3 groups are possible?

Solution 2.
There are

$$\frac{10!}{5!2!3!} = 2520 \quad \text{possible divisions.}$$

Exercise 3.
Prove that
$$\binom{n+m}{r} = \binom{n}{0}\binom{m}{r} + \binom{n}{1}\binom{m}{r-1} + \ldots + \binom{n}{r}\binom{m}{0}$$
Hint: From a box of n red balls and m blue balls, in how many ways you can select r balls from the box.

Solution 3.
This can be done in a combinatorial way by considering subsets of size r from a box of n red balls and m blue. The left hand side of the above represents one way of obtaining this identity. Another way to count the number of subsets of size r is to consider the number of possible sets can be found by considering a subproblem of how many red balls chosen to be included in the subset of size r. This number can range from zero to r red balls. When we have a subset of size r with zero red balls we must have all blue balls. This can be done in $\binom{n}{0}\binom{m}{r}$ ways. If we select one red ball and $r-1$ blus the number of subsets that meet this criterion is given by $\binom{n}{1}\binom{m}{r-1}$. Continuing this logic for all possible subset of the red balls we have the right hand side of the above expression.

Exercise 4.
Two dice are thrown. Let E be the event that the sum of the dice is odd, let F be the event that at least one of the dice lands on 1, and let G be the event that the sum is 5. Describe the events $E \cap F$, $E \cup F$, $F \cap G$, $E \cap F^c$ and $E \cap F \cap G$.

Solution 4.
Assume that the dice are not distinguishable, the events E, F, G are given by,

$$\begin{aligned}
E &= \{(1,2),(1,4),(1,6),(2,3),(2,5),(3,4),(3,6),(4,5),(5,6)\},\\
F &= \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)\},\\
G &= \{(1,4),(2,3)\}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
E \cap F &= \{(1,2),(1,4),(1,6)\},\\
E \cup F &= \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,3),(2,5),(3,4),(3,6),(4,5),(5,6)\},\\
F \cap G &= \{(1,4)\},\\
E \cap F^c &= \{(2,3),(2,5),(3,4),(3,6),(4,5),(5,6)\},\\
E \cap F \cap G &= \{(1,4)\}.
\end{aligned}$$

Exercise 5.
A classroom consists of 6 boys and 6 girls. What is the probability that if we randomly pick 3 students from the classroom, 2 of them are girls and 1 of them is a boy?

Solution 5.
Define $A := \{1 \text{ boy and } 2 \text{ girls}\}$, then the desired probability is given by

$$\begin{aligned}\mathbb{P}(A) &= \frac{\mathrm{C}_1^6 \mathrm{C}_2^6}{\mathrm{C}_3^{12}} \\ &= \frac{6 \times 15}{220} \\ &= \frac{9}{22}\end{aligned}$$

Exercise 6.
An instructor gives her class a set of 10 problems with the information that the final exam will consists of a random selection of 5 of them. If a student has figured out how to do 7 of problems, what is the probability that he or she will answer correctly

(a) all 5 problems?

(b) at least 4 of the problems?

Solution 6.
(a) Write

$$p_5 = \frac{\binom{7}{5}\binom{3}{0}}{\binom{10}{5}} = \frac{1}{12}.$$

(b) Similarly, we can find the probability that the student answer correctly 4 problems

$$p_4 = \frac{\binom{7}{4}\binom{3}{1}}{\binom{10}{5}} = \frac{5}{12},$$

hence, the desired probability is given by

$$p = \frac{5}{12} + \frac{1}{12} = \frac{1}{2}.$$

Exercise 7.
A bin contains 100 balls, of which 25 are red, 40 are white, and 35 are black. If two balls are selected from the bin without replacement, what is the probability that one will be red and one will be white?

Solution 7.
Let W stands for white and R for red

$$\begin{aligned}\mathbb{P}(RW) &= \frac{\binom{25}{1}\binom{40}{1}}{\binom{100}{2}} \\ &= \frac{20}{99}\end{aligned}$$